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Neural-network-based robust critic learning control with advanced value iteration for continuous-time dynamical systems

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ABSTRACT

In this paper, a novel neural-network-based robust critic learning method is developed with advanced value iteration (VI) to design the controller for continuous-time nonlinear systems. The initial admissible control law is eliminated by employing the developed scheme, which significantly relaxes the starting conditions for iteration. Additionally, the approximation error is analyzed to provide guidelines for parameter selection, thereby facilitating implementation of the constructed algorithm. By introducing a relaxation factor, the convergence process is accelerated leading to the better performance than the traditional VI algorithm. The convergence of the algorithm and the stability of the system are proved in detail. Finally, three examples are presented to verify the effectiveness of the established approach.

1. Introduction

Reinforcement learning (RL) is a powerful method that can be applied in many fields (Wang et al., 2023c; Yue et al., 2024). Based on the idea of trial-and-error learning, the policy that maximizes the reward can be found via the interaction between the agent and the environment (Wang et al., 2023b). Therefore, RL can be employed to solve the optimization problem (Wang et al., 2025). Actually, the optimization issue can be addressed by using dynamic programming (DP) when the model of the controlled system is known and simple (Kiumarsi et al., 2018). It can be regarded as the Markov decision process (MDP). The goal is to find a control policy that maximizes the total cumulative reward. According to this policy, an action is selected to the environment based on the observed current state. Then, the agent receives a reward that reflects the quality of the transition. Besides, the new state is collected and the cycle continues. However, “curse of dimensionality” appears when the number of elements in the state and action spaces is numerous in the MDP (Zhu & Zhao, 2018). Therefore, in this case, it is difficult to obtain the exact solution of the optimal control law (Wang et al., 2024a). To fill this gap, combined with RL, neural networks (NNs), and DP, adaptive dynamic programming (ADP) was proposed to deal with many kinds of optimal control problems for complex systems (Wu et al., 2023; Yang & He, 2020; Zhao et al., 2023). Its basic structure consists of the critic module, the model module, and the action module. These components

are typically implemented via using neural networks to handle the tasks of evaluation, prediction, and decision-making, respectively. ADP can be used to address many kinds of issues, such as the regulator problem, the tracking control problem, the multi-agent control problem, and the robust control problem for both discrete-time and continuous-time (CT) cases. Furthermore, in the real world, there are many large-scale systems. ADP can also be used to design whole control strategies for large-scale systems by transforming the decentralized control issue to the optimal control problem (Liu et al., 2024).

In real world, most controlled systems are always in the environment that contains all kinds of changes. To maintain the normal operation of the system, the robust control is proposed to deal with the uncertainty. There have been many methods to address the robust control problem such as sliding mode control, the linear matrix inequality, H_∞ control methods etc. Nevertheless, the model of the system is necessary for sliding mode control. Besides, the accuracy of the model has a great influence on the effect of the algorithm. The linear matrix inequality is invalid for the nonlinear system. None of these approaches takes the cost into consideration. In this paper, the idea of optimal-robust transformation is utilized to overcome issues mentioned above. The robust control problem is turned into the optimal control issue. Then, ADP schemes can be employed to solve the robust control problem.

Based on the idea of iteration, the near-optimal policy can be attained by utilizing ADP (Wang et al., 2023b). When conducting ADP

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(Wang et al., 2020), there are two common iteration methods: policy iteration (PI) (Peng et al., 2023) and value iteration (VI) (Ha et al., 2022). There have been many studies on the use of PI methods to control the continuous-time system (Liu et al., 2014; Xue et al., 2021, 2022). The properties of PI-based ADP algorithms have been analyzed, such as the convergence of the algorithm (Liu et al., 2014), the optimality of results (Xue et al., 2021), and the stability of system under the obtained controller (Xue et al., 2022). However, there is a consensus problem that researchers have acknowledged when using the PI approach (Yang et al., 2023). The initial control policy should be admissible, which is a key point in starting the iteration (Chen et al., 2024). Unfortunately, it is usually difficult to choose for real-world systems. Recently, there have been many studies on the subject of initial controller selection (Guo & Zhao, 2023; Lin et al., 2023; Xiao & Zhang, 2024; Xiao et al., 2018). Lin et al. (2023) used a new learning rule that contained an additional stability term, which worked when the system was in an unstable state. An extra NN was introduced to assist in attaining the optimal control law without requiring the system model and the initial admissible controller (Guo & Zhao, 2023). The core of many schemes is introducing other tools to compensate the process of finding the admissible control law. Hence, the application of the above methods is limited by the utilized tool. Besides, the computational complexity increases due to additional terms and NNs (Li et al., 2024; Wang et al., 2024b). To remove the premise of the initial admissible controller, based on the heuristic dynamic programming framework, VI-based ADP algorithms were employed to obtain the optimal control law for CT systems. Xiao and Zhang (2024) presented the convergence analysis of the VI-based ADP approach. Combining with the integral RL algorithm, VI was used to address the H_∞ control issue (Xiao et al., 2018). There is no requirement of the initial admissible control law. The condition of conducting VI is set as a random positive semidefinite function (Wang et al., 2023a), which is extremely relaxed compared to the initial condition for starting PI. However, the above methods are not available for evaluating the convergence rate of the algorithm. Ha et al. (2024) introduced the relaxation factor to accelerate the converge process when using VI. Combined with Q-learning, the accelerated VI method is used to address the safe tracking issue (Zhao et al., 2025). Li et al. (2025) applied accelerated VI to improve the convergence speed of optimal cooperative control scheme for the unknown linear discrete-time multi-agent systems. The structure of accelerated VI is effective in addressing many kinds of problems. However, works mentioned-above are focusing on the discrete-time nonlinear system. There is little research on VI-based accelerated ADP for CT systems.

Considering both the practicability and the convergence rate, a novel VI-based ADP method is developed for CT systems. Based on the structure of heuristic dynamic programming, we give the migration of VI to CT systems. The initial admissible control policy is eliminated by employing the VI algorithm. The speed of convergence process is enhanced on account of the introduction of the accelerated factor, which is rarely studied in VI-based ADP methods for CT systems. Then, by analyzing the approximation error, the selection of parameters is simplified when the developed scheme is carried out. Furthermore, similar to the literature mentioned above, the constructed method can also be used to design the controller for large-scale systems based on the idea of transforming to optimal control. Recently, deep HJB solvers like physics-informed neural networks (PINNs) have emerged, typically treating optimal control as a global residual minimization problem. However, PINNs are highly sensitive to hyperparameter tuning. Rigorous proofs also can not be provided when using PINNs. Compared with deep learning, the simpler NN is used in the developed scheme with less computational burden, which is suitable for many controlled systems. The main contributions are listed as follows.

1) To eliminate the initial admissible controller, the advanced VI-based ADP approach is introduced, which is first established to address the robust control problem for CT systems. The initial condition for implementing the iteration process is relaxed compared to the traditional

PI approach. Besides, no additional approximate tools are introduced to achieve this goal.

2) For CT nonlinear systems, an accelerated factor is added to the normal VI method to accelerate the convergence process. Compared with other VI-based methods for CT systems, the speed of the learning process is considered to reduce the time cost of the algorithm. Besides, the convergence of algorithm and the stability of system are analyzed in detail.

3) The approximation error is considered, which is generated from the approximation tool. The corresponding theorems are given to guide the selection of practical parameters, thereby facilitating the implementation of the developed algorithm.

Notations: $\mathbb{R}^n$ and $\mathbb{R}^{n \times m}$ represent the n -dimensional Euclidean space and the set of all real $n \times m$ matrices respectively. Ω is a compact subset of $\mathbb{R}^n$. $\Psi(\Omega)$ is the set of admissible control laws on Ω . $\|\cdot\|$ is the vector norm or matrix norm in $\mathbb{R}^n$ or $\mathbb{R}^{n \times m}$. T is a transposition symbol and $\nabla \triangleq \partial/\partial x$ denotes the gradient operator. I_n denotes the n -dimensional identity matrix. $\mathbb{N}^+$ denotes the set of positive integers.

The remainder of this paper is organized as follows. In Section 2, the robust control problem is described for nonlinear CT systems. In Section 3, the convergence of the algorithm and the stability of the system are presented in detail. In Section 4, we analyze the approximation error to facilitate the implementation of the developed method. In Section 5, three examples are given to verify the effectiveness of the advanced VI approach. In Section 6, we give the conclusion of this paper and future research directions.

2. Problem description

Uncertainties in system dynamics can lead to significant performance degradation, particularly in systems requiring high precision. By exploring robust control approaches, we seek to improve the resilience of the system against these disturbances.

Therefore, we consider a class of CT systems, which can be described as

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t) + \Delta f(x(t)), \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the system state, $f(x)$ and $g(x)$ are the system functions, and $u(t) \in \mathbb{R}^m$ is the control input. Assume $\Delta f(x(t)) = g(x)d(x)$ with $d(x) \in \mathbb{R}^m$, which is the matched uncertainty. For the bounded function $d(x)$ with $d(0) = 0$, we have $\|d(x)\| \leq d_M(x)$ with $d_M(0) = 0$. The basic condition is given in the following assumption.

Assumption 1. Suppose $f(x) \in \mathbb{R}^n$ and $g(x) \in \mathbb{R}^{n \times m}$ are locally Lipschitz continuous on a compact set $\Omega \subset \mathbb{R}^n$ with $f(0) = 0$.

In this paper, the idea of addressing the robust control problem is to convert it into the optimal control issue for the nominal system, which is written as

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t). \quad (2)$$

Therefore, the objective is transformed to acquire a control law that not only makes the system (2) stable but also minimizes the following cost function:

$$\begin{aligned} V(x_t) &= \int_t^\infty \{ \rho d_M^2(\tau) Q x(\tau) + u^\top(\tau) R u(\tau) \} d\tau \\ &= \int_t^\infty r(x(\tau), u(\tau)) d\tau, \end{aligned} \quad (3)$$

where ρ is a positive constant, $r(x(t), u(t))$ is the utility function, $x_t = x(t)$, $Q \in \mathbb{R}^{n \times n}$, and $R \in \mathbb{R}^{m \times m}$. If Assumption 1 is satisfied, then we have $V(x)$ is continuously differentiable.

The optimal cost function is defined as

$$V^*(x_t) = \min_{u \in \Psi(\Omega)} \int_t^\infty r(x(\tau), u(\tau)) d\tau. \quad (4)$$

If the control input is admissible and $V(x)$ is continuously differentiable, we have

$$0 = r(x, u) + (\nabla V(x))^T (f(x) + g(x)u), \quad (5)$$

where $\nabla V(x)$ denotes the partial differential of $V(x)$ with respect to the system state x . For simplicity, $x(t)$ is written as x .

The Hamiltonian is defined as

$$H(x, u(x), \nabla V(x)) = r(x, u(x)) + (\nabla V(x))^T (f(x) + g(x)u(x)). \quad (6)$$

Define the Hamilton-Jacobi-Bellman (HJB) equation as

$$0 = \min_{u \in \Psi(\Omega)} H(x, u(x), \nabla V^*(x)). \quad (7)$$

Then, we can deduce the optimal control law by solving the HJB equation, which can be described as

$$u^*(x) = -\frac{1}{2} R^{-1} g^T(x) \nabla V^*(x). \quad (8)$$

Remark 1: The equivalence of the robust and optimal control issue was given by Wang et al. (2024a). Therefore, it is not described in this paper.

3. Properties of the advanced VI-based ADP algorithm

In this section, traditional VI and PI are revisited to show the superiority of the developed method. Then, by introducing the relaxation factor w , the advanced VI-based ADP method is established for CT systems. Besides, the convergence of the developed algorithm is proved and the stability of the system is also analyzed when using the obtained policy based on the novel method.

3.1. Revisiting traditional PI and VI

Denote the iteration sequence $\{V_i(x)\}$ and $\{u_i(x)\}$. For the traditional PI scheme, the process of acquiring the optimal value function and policy are described below. First, we select an admissible control policy u_0 . Then, policy evaluation is conducted according to

$$0 = r(x, u_i(x)) + (\nabla V^{u_i}(x))^T (f(x) + g(x)u_i(x)), \quad (9)$$

where $V^{u_i}(x)$ denotes the value function related to $u_i(x)$ and $i = 0, 1, 2, \dots$. We implement policy improvement based on

$$\begin{aligned} u_{i+1} &= \arg \min_{u \in \Psi(\Omega)} [H(x, u, \nabla V^{u_i})] \\ &= -\frac{1}{2} R^{-1} g^T(x) \nabla V^{u_i}(x). \end{aligned} \quad (10)$$

The traditional VI algorithm can be initialized by setting $V_0(x)$ as a positive semi-definite function and presented as the value function update:

$$V_i(x) = \int_t^{t+T} r(x, u_{i-1}(x)) d\tau + V_{i-1}(X), \quad (11)$$

where $X = x + \int_t^{t+T} \{f(x) + g(x)u\} d\tau$ and $i = 1, 2, \dots$ and policy improvement:

$$u_i(x) = \arg \min_u \left[\int_t^{t+T} r(x, u(x)) d\tau + V_i(X) \right]. \quad (12)$$

3.2. Advanced VI-based ADP

By revisiting PI and VI, the properties of the traditional method are shown to highlight the advantages of the developed scheme. Unlike PI, the initial admissible control law is not required in advanced VI-based ADP. Compared to VI, the learning process is improved in the established approach. The iterative procedure is performed between the novel value function update:

$$\tilde{V}_i(x) = w \left(\int_t^{t+T} r(x, \tilde{u}_{i-1}) d\tau + \tilde{V}_{i-1}(X) \right) + (1-w) \tilde{V}_{i-1}(x) w \quad (13)$$

and novel policy improvement:

$$\tilde{u}_i(x) = \arg \min_{\tilde{u}} \left[\int_t^{t+T} r(x, \tilde{u}(x)) d\tau + \tilde{V}_i(X) \right]. \quad (14)$$

Assumption 2. Suppose that the conditions $0 \leq V^*(X) \leq K \int_t^{t+T} r(x, \tilde{u}) d\tau$ and $0 \leq \underline{\lambda} V^*(x) \leq \tilde{V}_0(x) \leq \bar{\lambda} V^*(x)$ are satisfied, where $0 < K < \infty$ and $0 \leq \underline{\lambda} \leq 1 \leq \bar{\lambda}$.

Theorem 1. Let the value function and the control policy be iteratively updated by (13) and (14). If Assumption 2 and $0 < w \leq 1$ hold, then the iterative value function $\tilde{V}_i(x)$ approximates the optimal value function $V^*(x)$ according to the following inequalities:

$$\begin{aligned} \left[1 - \left(1 - \frac{w}{1+K} \right)^i (1 - \underline{\lambda}) \right] V^*(x) &\leq \tilde{V}_i(x) \\ &\leq \left[1 + \left(1 - \frac{w}{1+K} \right)^i (\bar{\lambda} - 1) \right] V^*(x). \end{aligned} \quad (15)$$

Proof: Denote $U(x, \tilde{u}) = \int_t^{t+T} r(x, \tilde{u}) d\tau$. Considering $0 \leq \underline{\lambda} V^*(x) \leq \tilde{V}_0(x)$ described in Assumption 2, it can be deduced that

$$\begin{aligned} \tilde{V}_1(x) &= w \min_{\tilde{u}} \{ U(x, \tilde{u}) + \tilde{V}_0(X) \} \\ &\quad + (1-w) \tilde{V}_0(x) \\ &\geq w \min_{\tilde{u}} \{ U(x, \tilde{u}) + \underline{\lambda} V^*(X) \} \\ &\quad + (1-w) \underline{\lambda} V^*(x). \end{aligned} \quad (16)$$

Based on $0 \leq V^*(X) \leq K \int_t^{t+T} r(x, u) d\tau$ in Assumption 2 and (16), it can be obtained that

$$\begin{aligned} \tilde{V}_1(x) &\geq w \min_{\tilde{u}} \left\{ \left(1 - K \frac{1-\underline{\lambda}}{1+K} \right) U(x, \tilde{u}) + \left(\underline{\lambda} + \frac{1-\underline{\lambda}}{1+K} \right) V^*(X) \right\} \\ &\quad + (1-w) \underline{\lambda} V^*(x) \\ &= \left[w \frac{\underline{\lambda} K + 1}{1+K} + (1-w) \underline{\lambda} \right] V^*(x) \\ &= \left[\left(1 - \frac{w}{1+K} \right) \underline{\lambda} + \frac{w}{1+K} \right] V^*(x). \end{aligned} \quad (17)$$

To derive the inequality corresponding to $\tilde{V}_i(x)$, the above process is conducted for $i-1$ times, which can be written as

$$\begin{aligned} \tilde{V}_i(x) &\geq \left[\left(1 - \frac{w}{1+K} \right)^i \underline{\lambda} + \left(1 - \frac{w}{1+K} \right)^{i-1} \frac{w}{1+K} \right. \\ &\quad \left. + \dots + \left(1 - \frac{w}{1+K} \right) \frac{w}{1+K} + \frac{w}{1+K} \right] V^*(x) \\ &= \left[\left(1 - \frac{w}{1+K} \right)^i \underline{\lambda} + \sum_{p=0}^{i-1} \frac{w}{1+K} \left(1 - \frac{w}{1+K} \right)^p \right] \\ &\quad \times V^*(x) \\ &= \left[\left(1 - \frac{w}{1+K} \right)^i \underline{\lambda} + 1 - \left(1 - \frac{w}{1+K} \right)^i \right] V^*(x). \end{aligned} \quad (18)$$

Therefore, the left-hand side of (15) is given. Similar to the proof process of the left-hand side, the right-hand side can be derived according to $\tilde{V}_0(x) \leq \bar{\lambda} V^*(x)$ in Assumption 2. Therefore, when i tends to infinity, we can obtain

$$\lim_{i \rightarrow \infty} \left[1 - \left(1 - \frac{w}{1+K} \right)^i (1 - \underline{\lambda}) \right] V^*(x) = V^*(x) \quad (19)$$

and

$$\lim_{i \rightarrow \infty} \left[1 + \left(1 - \frac{w}{1+K} \right)^i (\bar{\lambda} - 1) \right] V^*(x) = V^*(x). \quad (20)$$

The proof is completed.

According to the above proof, the convergence of the developed algorithm is presented for $0 < w \leq 1$. Then, the following theorem is given to discuss the situation of $w > 1$.

Theorem 2. Suppose that Assumption 2 holds. Let the value function and the control policy be iteratively updated by (13) and (14). If the following condition is satisfied:

$$1 \leq w \leq 1 + \frac{\mathcal{L}\mathcal{D}_{\min}}{(1+K)(\bar{\lambda}-\underline{\lambda})}, \quad (21)$$

where $\mathcal{L} \in (0, 1)$ and $\mathcal{D}_{\min} = \min\{1 - \underline{\lambda}, \bar{\lambda} - 1\}$, then $\tilde{V}_i(x)$ approximates $V^*(x)$ in the following manner:

$$\begin{aligned} \left[1 - \left(1 - \frac{w - \mathcal{L}}{1+K}\right)^i (1 - \underline{\lambda})\right] V^*(x) &\leq \tilde{V}_i(x) \\ &\leq \left[1 + \left(1 - \frac{w - \mathcal{L}}{1+K}\right)^i (\bar{\lambda} - 1)\right] V^*(x). \end{aligned} \quad (22)$$

Proof: According to the condition (21), we can deduce that

$$w \leq 1 + \frac{\mathcal{L}(1 - \underline{\lambda})}{(1+K)(\bar{\lambda} - \underline{\lambda})} \quad (23)$$

and

$$w \leq 1 + \frac{\mathcal{L}(\bar{\lambda} - 1)}{(1+K)(\bar{\lambda} - \underline{\lambda})}. \quad (24)$$

It can be further written as

$$(1 - w)\bar{\lambda} \geq (1 - w)\underline{\lambda} + \frac{\mathcal{L}(\bar{\lambda} - 1)}{1 + K} \quad (25)$$

and

$$(1 - w)\underline{\lambda} \leq (1 - w)\bar{\lambda} + \frac{\mathcal{L}(\bar{\lambda} - 1)}{1 + K}. \quad (26)$$

Considering the above derivation and Assumption 2, we have the following relationship based on the idea of proving Theorem 1:

$$\begin{aligned} \tilde{V}_i(x) &\leq w \min_{\tilde{u}} \left\{ \left(1 + K \frac{\bar{\lambda} - 1}{1 + K}\right) U(x, \tilde{u}) + \left(\bar{\lambda} - \frac{\bar{\lambda} - 1}{1 + K}\right) V^*(x) \right\} \\ &\quad + (1 - w)\underline{\lambda} V^*(x) \\ &= \left[w \frac{\bar{\lambda} K + 1}{1 + K} + (1 - w)\underline{\lambda} \right] V^*(x) \\ &\leq \left[\left(1 - \frac{w - \mathcal{L}}{1 + K}\right) \bar{\lambda} + \frac{w - \mathcal{L}}{1 + K} \right] V^*(x). \end{aligned} \quad (27)$$

Similar to the proof of the upper bound of $\tilde{V}_i(x)$, the lower bound can be obtained leading to

$$\begin{aligned} \left[1 - \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i (1 - \underline{\lambda})\right] V^*(x) &\leq \tilde{V}_i(x) \\ &\leq \left[1 + \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i (\bar{\lambda} - 1)\right] V^*(x). \end{aligned} \quad (28)$$

Define $\underline{\lambda}_1 = 1 - (1 - ((w - \mathcal{L})/(1 + K)))(1 - \underline{\lambda})$ and $\bar{\lambda}_1 = 1 + (1 - ((w - \mathcal{L})/(1 + K)))(\bar{\lambda} - 1)$, where $0 \leq \underline{\lambda}_1 \leq 1 \leq \bar{\lambda}_1$. Then, we have

$$\begin{aligned} w &\leq 1 + \frac{\mathcal{L}(1 - \underline{\lambda})}{(1+K)(\bar{\lambda} - \underline{\lambda})} \\ &= 1 + \frac{\left(1 - \frac{w - \mathcal{L}}{1+K}\right)(1 - \underline{\lambda})}{(1+K)\left(1 - \frac{w - \mathcal{L}}{1+K}\right)(\bar{\lambda} - \underline{\lambda})} \mathcal{L} \\ &= 1 + \frac{\mathcal{L}(1 - \underline{\lambda}_1)}{(1+K)(\bar{\lambda}_1 - \underline{\lambda}_1)} \end{aligned} \quad (29)$$

and

$$\begin{aligned} w &\leq 1 + \frac{\mathcal{L}(\bar{\lambda} - 1)}{(1+K)(\bar{\lambda} - \underline{\lambda})} \\ &= 1 + \frac{\left(1 - \frac{w - \mathcal{L}}{1+K}\right)(\bar{\lambda}_1 - 1)}{(1+K)\left(1 - \frac{w - \mathcal{L}}{1+K}\right)(\bar{\lambda}_1 - \underline{\lambda}_1)} \mathcal{L} \end{aligned}$$

$$= 1 + \frac{\mathcal{L}(\bar{\lambda}_1 - 1)}{(1+K)(\bar{\lambda}_1 - \underline{\lambda}_1)}. \quad (30)$$

Define $\underline{\lambda}_i = 1 - (1 - ((w - \mathcal{L})/(1 + K)))^i (1 - \underline{\lambda})$ and $\bar{\lambda}_i = 1 + (1 - ((w - \mathcal{L})/(1 + K)))^i (\bar{\lambda} - 1)$. Therefore, we have

$$\begin{aligned} \tilde{V}_i(x) &\geq w \min_{\tilde{u}} \left\{ \left(1 - K \frac{1 - \underline{\lambda}_{i-1}}{1 + K}\right) U(x, \tilde{u}) + \left(\underline{\lambda}_{i-1} + \frac{1 - \underline{\lambda}_{i-1}}{1 + K}\right) V^*(x) \right\} \\ &\quad + (1 - w)\bar{\lambda}_{i-1} V^*(x) \\ &\geq \left[\left(1 - \frac{w - \mathcal{L}}{1 + K}\right) \underline{\lambda}_{i-1} + \frac{w - \mathcal{L}}{1 + K} \right] V^*(x). \end{aligned} \quad (31)$$

Expanding to the i th iteration, it can be obtained as

$$\begin{aligned} \tilde{V}_i(x) &\geq \left[\left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i \underline{\lambda} + \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^{i-1} \frac{w - \mathcal{L}}{1 + K} \right. \\ &\quad \left. + \dots + \left(1 - \frac{w - \mathcal{L}}{1 + K}\right) \frac{w - \mathcal{L}}{1 + K} + \frac{w - \mathcal{L}}{1 + K} \right] V^*(x) \\ &= \left[\left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i \underline{\lambda} + \sum_{p=0}^{i-1} \frac{w - \mathcal{L}}{1 + K} \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^p \right] \\ &\quad \times V^*(x) \\ &= \left[\left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i \underline{\lambda} + 1 - \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i \right] V^*(x). \end{aligned} \quad (32)$$

Analyzing the condition (21), we can derive that

$$1 \leq w \leq 1 + \frac{\mathcal{L}}{(1+K)(\bar{\lambda} - \underline{\lambda})} \leq 1 + \mathcal{L}, \quad (33)$$

leading to

$$0 \leq \frac{w - \mathcal{L}}{1 + K} \leq \frac{1 + \mathcal{L} + K - \mathcal{L}}{1 + K} = 1. \quad (34)$$

It can be concluded that when $i \rightarrow \infty$, we have

$$\lim_{i \rightarrow \infty} \left[1 - \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i (1 - \underline{\lambda}) \right] V^*(x) = V^*(x) \quad (35)$$

and

$$\lim_{i \rightarrow \infty} \left[1 + \left(1 - \frac{w - \mathcal{L}}{1 + K}\right)^i (\bar{\lambda} - 1) \right] V^*(x) = V^*(x). \quad (36)$$

The proof is completed.

According to the above description, the convergence of the developed algorithm is proved for two situations about the value of w . Then, Theorem 3 is given to prove that the iterative value function is positive definite.

Theorem 3. Suppose Assumption 2 holds. Let the value function and the control policy be iteratively updated by (13) and (14). If the following condition is satisfied:

$$0 < w \leq 1 + \frac{\bar{\lambda} - \underline{\lambda} - \mathcal{D}_{\max}}{K(\bar{\lambda} - \underline{\lambda}) + \mathcal{D}_{\max}}, \quad (37)$$

where $\mathcal{D}_{\max} = \max\{\bar{\lambda} - 1, 1 - \underline{\lambda}\}$, then $\tilde{V}_i$ is positive definite.

Proof: First, observing the Eq. (15), we can find $\tilde{V}_i$ is positive definite when $0 < w \leq 1$. Then, we consider the situation of $1 < w$. By rewriting the right-hand side of (37), the following condition is discussed:

$$1 < w \leq \frac{K(\bar{\lambda} - \underline{\lambda}) + \bar{\lambda} - \underline{\lambda}}{K(\bar{\lambda} - \underline{\lambda}) + \mathcal{D}_{\max}}. \quad (38)$$

Meanwhile, we have

$$w \leq \frac{(K + 1)(\bar{\lambda} - \underline{\lambda})}{K(\bar{\lambda} - \underline{\lambda}) + \bar{\lambda} - 1} \quad (39)$$

and

$$w \leq \frac{(K+1)(\bar{\lambda} - \underline{\lambda})}{K(\bar{\lambda} - \underline{\lambda}) + 1 - \underline{\lambda}}. \quad (40)$$

Due to $K(\bar{\lambda} - \underline{\lambda}) + \mathcal{D}_{\max} > 0$ and $1 + K > 0$, inequalities (39) and (40) can be rewritten as

$$w \frac{\lambda K + 1}{1 + K} - w \bar{\lambda} = \frac{K(\bar{\lambda} - \underline{\lambda}) + 1 - \bar{\lambda}}{1 + K} w \geq \underline{\lambda} - \bar{\lambda} \quad (41)$$

and

$$w \frac{\bar{\lambda} K + 1}{1 + K} - w \underline{\lambda} = \frac{K(\bar{\lambda} - \underline{\lambda}) + 1 - \underline{\lambda}}{1 + K} w \leq \bar{\lambda} - \underline{\lambda}. \quad (42)$$

Considering (41) and $\tilde{V}_1(x) \geq [w((\lambda K + 1)/(1 + K)) + (1 - w)\bar{\lambda}]V^*(x)$, we obtain

$$\tilde{V}_1(x) \geq \underline{\lambda} V^*(x) \quad (43)$$

and

$$\tilde{V}_1(x) \leq \bar{\lambda} V^*(x). \quad (44)$$

Based on (43) and (44), it can be deduced that $0 \leq \underline{\lambda} V^*(x) \leq \tilde{V}_1(x) \leq \bar{\lambda} V^*(x)$. It can be concluded that $0 \leq \underline{\lambda} V^*(x) \leq \tilde{V}_i(x) \leq \bar{\lambda} V^*(x)$ holds, which completes the proof.

Assumption 2 and other conditions are set to prove the convergence of the developed scheme. In **Assumption 2**, the initial condition $0 \leq \underline{\lambda} V^*(x) \leq \tilde{V}_0(x) \leq \bar{\lambda} V^*(x)$ is easy to achieve. From the aspect of intuitive explanation, the condition $0 \leq V^*(X) \leq K \int_t^{t+T} r(x, \tilde{u}) d\tau$ can be explained as the optimal cost function $V^*(X)$ is bounded above by K times the cost integrated over a fixed time interval T . It implies that the optimal cost function can be learned in a finite time.

Next, we discuss the stability of the system by using the obtained control policy based on the constructed method.

Theorem 4. Let the value function and the control policy be iteratively updated by (13) and (14). Assume $\tilde{V}_0(x) = \int_t^{t+T} \{\rho d_M^2 + Q(x)\} d\tau = Q(x)$. For any $x \neq 0$, the following condition is satisfied:

$$\tilde{V}_{i+1}(x) - \tilde{V}_i(x) < w c \int_t^{t+T} \{\rho d_M^2 + Q(x)\} d\tau, \quad (45)$$

where w satisfies the condition (37) and $c \in (0, 1)$. Then, the control policy $\tilde{u}_i(x)$ is admissible.

Proof: According to the Eq. (13), for any $x \neq 0$, (45) can be rewritten as

$$w \left[U(x, \tilde{u}_i(x)) + \tilde{V}_i(x) - \tilde{V}_i(x) \right] < w c Q(x). \quad (46)$$

Furthermore, it can be presented as

$$\tilde{V}_i(x) - \tilde{V}_i(x) < c Q(x) - U(x, \tilde{u}_i(x)) < 0. \quad (47)$$

According to **Theorem 3**, $\tilde{V}_i(x)$ is positive definite. By selecting $\tilde{V}_i(x)$ as the Lyapunov function, it can be deduced that $\tilde{u}_i(x)$ can stabilize the closed-loop system based on (47).

For any $x \in \Omega$, we can obtain that

$$\tilde{V}_i(x(t+T)) - \tilde{V}_i(x(t)) \leq (c-1)Q(x(t)) - \mathcal{R}(x(t), \tilde{u}_i), \quad (48)$$

where $\mathcal{R}(x, \tilde{u}_i) = \int_t^{t+T} \tilde{u}_i^T R \tilde{u}_i d\tau$.

For $x(t+2T)$, $x(t+3T)$, ..., $x(t+jT)$, the following inequalities are presented as:

$$\begin{aligned} & \tilde{V}_i(x(t+2T)) - \tilde{V}_i(x(t+T)) \\ & \leq (c-1)Q(x(t+T)) - \mathcal{R}(x(t+T), \tilde{u}_i) \\ & \tilde{V}_i(x(t+3T)) - \tilde{V}_i(x(t+2T)) \\ & \leq (c-1)Q(x(t+2T)) - \mathcal{R}(x(t+2T), \tilde{u}_i) \\ & \vdots \\ & \tilde{V}_i(x(t+jT)) - \tilde{V}_i(x(t+(j-1)T)) \end{aligned} \quad (49)$$

$$\begin{aligned} & \leq (c-1)Q(x(t+(j-1)T)) \\ & - \mathcal{R}(x(t+(j-1)T), \tilde{u}_i), \end{aligned} \quad (50)$$

where $j \in \mathbb{N}^+$. The above inequalities can be expressed similarly when $j \rightarrow \infty$. Due to the asymptotic stability by using $\tilde{u}_i(x)$, we have

$$\lim_{j \rightarrow \infty} \tilde{V}_i(x(t+jT)) = 0. \quad (51)$$

Considering (48)–(51), it can be concluded that

$$\begin{aligned} & \tilde{V}_i(x(t)) \\ & \geq \sum_{j=0}^{\infty} [(1-c)Q(x(t+jT)) + \mathcal{R}(x(t+jT), \tilde{u}_i)] \\ & \geq (1-c) \sum_{j=0}^{\infty} [Q(x(t+jT)) + \mathcal{R}(x(t+jT), \tilde{u}_i)], \end{aligned} \quad (52)$$

where $\mathcal{R}(x(t+jT), \tilde{u}_i) = \int_{t+jT}^{t+(j+1)T} \tilde{u}_i^T R \tilde{u}_i d\tau$. It implies that

$$\int_t^{\infty} \{\rho d_M^2 + Q(x) + \tilde{u}_i^T R \tilde{u}_i\} d\tau \leq \frac{1}{1-c} \tilde{V}_i(x(t)). \quad (53)$$

Due to $\tilde{V}_i(x(t))$ is finite, which is proved in **Theorem 1**, the cost function is finite, which implies $\tilde{u}_i(x)$ is admissible. The proof is completed.

4. Analysis of the approximation error

Convergence of the algorithm and system stability are given according to the above proof. However, to implement the scheme, approximating tools should be used to approximate the value function. In this paper, a critic NN is used to estimate the value function, defined as

$$\hat{\tilde{V}}_i(x) = \hat{\omega}_c^T \sigma_c(x), \quad (54)$$

where $\hat{\omega}_c \in \mathbb{R}^l$ is the weight vector of the NN and $\sigma_c(x)$ is the activation function with $\sigma_c(0) = 0$. l denotes the number of hidden-layer neurons in the critic NN. Then, the related control law can be written as

$$\hat{\tilde{u}}_i(x) = -\frac{1}{2} R^{-1} g^T(x) (\nabla \sigma_c(x))^T \hat{\omega}_c. \quad (55)$$

The error between the estimated value function and the actual one is presented as

$$e_c = \hat{\omega}_c^T \sigma_c(x) - \tilde{V}_i(x). \quad (56)$$

The objective function is defined as $E_c = (1/2)e_c^T e_c$. The critic NN is trained to minimize E_c . Due to the introduction of the neural network, the approximate error is added when conducting the developed scheme. Although the approximate error can be proved to be uniformly ultimately bounded by Wang et al. (2017), it still influences the success rate of algorithm implementation. To facilitate the implementation of the advanced VI-based ADP scheme, the approximation error is analyzed by presenting the following theorems.

Theorem 5. Suppose **Assumption 2** is satisfied and $0 < w \leq 1$ holds. Let the value function and the control policy be iteratively updated by (13) and (14). Assume $0 < \delta < 1 < \bar{\delta}$. Considering the approximation error, if the inequality related to the upper bound of the approximation factor δ holds, i.e., $\bar{\delta} < (1+K)/(1+K-w)$ holds, we have

$$\underline{\sigma}_i V^*(x) \leq \hat{\tilde{V}}_i(x) \leq \bar{\sigma}_i V^*(x), \quad (57)$$

where

$$\underline{\sigma}_i = \left(1 - \frac{w}{1+K}\right)^i \delta^i \underline{\lambda} + \sum_{p=0}^{i-1} \frac{w \delta^{p+1}}{1+K} \left(1 - \frac{w}{1+K}\right)^p \quad (58)$$

and

$$\bar{\sigma}_i = \left(1 - \frac{w}{1+K}\right)^i \bar{\delta}^i \bar{\lambda} + \sum_{p=0}^{i-1} \frac{w \bar{\delta}^{p+1}}{1+K} \left(1 - \frac{w}{1+K}\right)^p. \quad (59)$$

Proof: Based on $\underline{\lambda}V^*(x) \leq \tilde{V}_0(x) = \hat{V}_0(x) \leq \bar{\lambda}V^*(x)$, we have

$$\begin{aligned} \hat{V}_1(x(t)) &\geq \underline{\delta}w \min_{\bar{u}} \left[\int_t^{t+T} r(x, \bar{u}) d\tau + \hat{V}_0(X) \right] \\ &\quad + \underline{\delta}(1-w)\hat{V}_0(x) \\ &\geq \underline{\delta}w \min_{\bar{u}} \left[\int_t^{t+T} r(x, \bar{u}) d\tau + \underline{\lambda}V^*(X) \right] \\ &\quad + \underline{\delta}(1-w)\underline{\lambda}V^*(x) \\ &\geq \underline{\delta}w \min_{\bar{u}} \left[\left(1 - K \frac{1-\underline{\lambda}}{1+K}\right) \int_t^{t+T} r(x, \bar{u}) d\tau \right. \\ &\quad \left. + \left(\underline{\lambda} + \frac{1-\underline{\lambda}}{1+K}\right)V^*(X) \right] + \underline{\delta}(1-w)\underline{\lambda}V^*(x) \\ &= \left(\left(\underline{\delta} - \frac{\underline{\delta}w}{1+K}\right)\underline{\lambda} + \frac{\underline{\delta}w}{1+K} \right) V^*(x) \\ &= \underline{\delta}\underline{\lambda}_1 V^*(x), \end{aligned} \quad (60)$$

where $\underline{\delta}$ is the lower bound of δ and $\underline{\lambda}_1 = (1 - (w/(1+K)))\underline{\lambda} + (w/(1+K))$.

Therefore, we have

$$\begin{aligned} \hat{V}_i(x) &\geq \left[\left(1 - \frac{w}{1+K}\right)^i \underline{\delta}^i \underline{\lambda} + \left(1 - \frac{w}{1+K}\right)^{i-1} \frac{w\underline{\delta}^i}{1+K} \right. \\ &\quad \left. + \dots + \left(1 - \frac{w}{1+K}\right) \frac{w\underline{\delta}^2}{1+K} + \frac{w\underline{\delta}}{1+K} \right] V^*(x) \\ &= \left[\left(1 - \frac{w}{1+K}\right)^i \underline{\delta}^i \underline{\lambda} + \sum_{p=0}^{i-1} \frac{w\underline{\delta}^{p+1}}{1+K} \left(1 - \frac{w}{1+K}\right)^p \right] \\ &\quad \times V^*(x) \\ &= \underline{\sigma}_i V^*(x). \end{aligned} \quad (61)$$

When $i \rightarrow \infty$, it can be deduced that

$$0 < \lim_{i \rightarrow \infty} \underline{\sigma}_i = \frac{w\underline{\delta}}{(1+K)(1-\underline{\delta}) + w\underline{\delta}} < 1. \quad (62)$$

For the upper bound of (57), by employing the idea of proof utilized above, we have

$$\begin{aligned} \hat{V}_i(x) &\leq \left[\left(1 - \frac{w}{1+K}\right)^i \bar{\delta}^i \bar{\lambda} + \left(1 - \frac{w}{1+K}\right)^{i-1} \frac{w\bar{\delta}^i}{1+K} \right. \\ &\quad \left. + \dots + \left(1 - \frac{w}{1+K}\right) \frac{w\bar{\delta}^2}{1+K} + \frac{w\bar{\delta}}{1+K} \right] V^*(x) \\ &= \left[\left(1 - \frac{w}{1+K}\right)^i \bar{\delta}^i \bar{\lambda} + \sum_{p=0}^{i-1} \frac{w\bar{\delta}^{p+1}}{1+K} \left(1 - \frac{w}{1+K}\right)^p \right] \\ &\quad \times V^*(x) \\ &= \bar{\sigma}_i V^*(x). \end{aligned} \quad (63)$$

If the following condition is satisfied:

$$\bar{\delta} < \frac{1+K}{1+K-w}, \quad (64)$$

then

$$\lim_{i \rightarrow \infty} \bar{\sigma}_i = \frac{w\bar{\delta}}{1+K-\bar{\delta}(1+K-w)} > 1. \quad (65)$$

The proof is completed.

For the analysis of the approximation error, the situation of $0 < w \leq 1$ is given as the proof mentioned above. According to Theorem 5, it can be found that if the upper bound of δ is set to meet the condition, then

sequence of $\hat{V}_i(x)$ converge in the manner of (56) when $0 < w \leq 1$. It reveals that the significance of Theorem 5, which guarantees the convergence of the value function and provides the guidance for choosing the approximation factor. Next, the discussion of $w > 1$ is presented in the following description.

Similar to the condition (21), Assumption 3 is set to prove the convergence of the developed scheme and provide the relationship between the error and the accelerated factor.

Assumption 3. Suppose that

$$w \leq 1 + \frac{\mathcal{L}(1-\underline{a})}{(1+K)(\bar{\lambda}-\underline{\lambda})} \quad (66)$$

and

$$w \leq 1 + \frac{\mathcal{L}(\bar{a}-1)}{(1+K)(\bar{\lambda}-\underline{\lambda})}, \quad (67)$$

where

$$\underline{a} = \left(1 - \frac{w-\mathcal{L}}{1+K}\right)\underline{\lambda} + \frac{w-\mathcal{L}\bar{\delta}}{1+K} \quad (68)$$

and

$$\bar{a} = \left(1 - \frac{w-\mathcal{L}}{1+K}\right)\bar{\lambda} + \frac{w-\mathcal{L}\bar{\delta}}{1+K}. \quad (69)$$

Theorem 6. Suppose Assumptions 2 and 3 are satisfied. Let the value function and the control policy be iteratively updated by (13) and (14). Considering the approximation error, when $i > 1$, if $w < 1+K$ and $\bar{\delta} < (1+K)/(1+K-w+\mathcal{L})$ hold, then

$$\underline{\sigma}_i V^*(x) \leq \hat{V}_i(x) \leq \bar{\sigma}_i V^*(x), \quad (70)$$

where

$$\begin{aligned} \underline{\sigma}_i &= \underline{\delta}^i \left(1 - \frac{w-\mathcal{L}}{1+K}\right)^i \underline{\lambda} + \underline{\delta}^i \left(1 - \frac{w-\mathcal{L}}{1+K}\right)^{i-1} \frac{w-\mathcal{L}}{1+K} \\ &\quad + \dots + \sum_{p=0}^{i-2} \frac{w-\mathcal{L}\bar{\delta}}{1+K} \underline{\delta}^{p+1} \left(1 - \frac{w-\mathcal{L}}{1+K}\right)^p \end{aligned} \quad (71)$$

and

$$\begin{aligned} \bar{\sigma}_i &= \bar{\delta}^i \left(1 - \frac{w-\mathcal{L}}{1+K}\right)^i \bar{\lambda} + \bar{\delta}^i \left(1 - \frac{w-\mathcal{L}}{1+K}\right)^{i-1} \frac{w-\mathcal{L}}{1+K} \\ &\quad + \dots + \sum_{p=0}^{i-2} \frac{w-\mathcal{L}\bar{\delta}}{1+K} \bar{\delta}^{p+1} \left(1 - \frac{w-\mathcal{L}}{1+K}\right)^p. \end{aligned} \quad (72)$$

Proof: Similar to the situation of $0 < w \leq 1$, the inequality related to $\hat{V}_1(x)$ can be presented as

$$\begin{aligned} \hat{V}_1(x) &\geq w\underline{\delta} \min_{\bar{u}} \left\{ \left(1 + K \frac{\underline{\lambda}-1}{1+K}\right) U(x, \bar{u}) \right. \\ &\quad \left. + \left(\underline{\lambda} - \frac{\underline{\lambda}-1}{1+K}\right) V^*(X) \right\} \\ &\quad + (1-w)\underline{\delta}\bar{\lambda}V^*(x) \\ &= \underline{\delta} \left[w \frac{\underline{\lambda}K+1}{1+K} + (1-w)\bar{\lambda} \right] V^*(x) \\ &\geq \underline{\delta} \left[\left(1 - \frac{w-\mathcal{L}}{1+K}\right)\underline{\lambda} + \frac{w-\mathcal{L}}{1+K} \right] V^*(x). \end{aligned} \quad (73)$$

Let $\underline{\lambda}_1 = (1 - ((w-\mathcal{L})/(1+K)))\underline{\lambda} + (w-\mathcal{L})/(1+K)$ and $\bar{\lambda}_1 = (1 - ((w-\mathcal{L})/(1+K)))\bar{\lambda} + (w-\mathcal{L})/(1+K)$. Then, we can deduce that $0 \leq \underline{\lambda}_1 \leq 1 \leq \bar{\lambda}_1$. According to (23) and (24), we have (29) and (30). Hence, the same idea can be used to derive the inequality of $\hat{V}_2(x)$, which can be written as

$$\begin{aligned} \hat{V}_2(x) &\geq w\underline{\delta} \min_{\bar{u}} \left\{ \left(1 + K \frac{\underline{\lambda}_1\bar{\delta}-1}{1+K}\right) U(x, \bar{u}) + \left(\underline{\lambda}_1\bar{\delta} - \frac{\underline{\lambda}_1\bar{\delta}-1}{1+K}\right) V^*(X) \right\} \\ &\quad + (1-w)\underline{\delta}^2\bar{\lambda}_1 V^*(x) \end{aligned}$$

$$\begin{aligned}
&= \left[w \underline{\delta} \frac{\lambda_1 \underline{\delta} K + 1}{1 + K} + (1 - w) \underline{\delta}^2 \bar{\lambda}_1 \right] V^*(x) \\
&\geq \underline{\delta} \left[\left(1 - \frac{w - \mathcal{L}}{1 + K} \right) \underline{\delta} \lambda_1 + \frac{w - \mathcal{L} \underline{\delta}}{1 + K} \right] V^*(x). \quad (74)
\end{aligned}$$

Let $\lambda_2 = (1 - ((w - \mathcal{L})/(1 + K)))\lambda_1 + (w - \mathcal{L}\underline{\delta})/(1 + K)$. Based on [Assumption 3](#), inequalities (23) and (24) hold. Therefore, when $i > 1$, it can be deduced that

$$\begin{aligned}
&\hat{V}_i(x) \\
&\geq \left[\underline{\delta}^i \left(1 - \frac{w - \mathcal{L}}{1 + K} \right)^i \lambda + \underline{\delta}^i \left(1 - \frac{w - \mathcal{L}}{1 + K} \right)^{i-1} \frac{w - \mathcal{L}}{1 + K} \right. \\
&\quad \left. + \dots + \underline{\delta}^2 \left(1 - \frac{w - \mathcal{L}}{1 + K} \right) \frac{w - \mathcal{L} \underline{\delta}}{1 + K} + \frac{w - \mathcal{L} \underline{\delta}^2}{1 + K} \right] V^*(x) \\
&= \left[\underline{\delta}^i \left(1 - \frac{w - \mathcal{L}}{1 + K} \right)^i \lambda + \underline{\delta}^i \left(1 - \frac{w - \mathcal{L}}{1 + K} \right)^{i-1} \frac{w - \mathcal{L}}{1 + K} \right. \\
&\quad \left. + \dots + \sum_{p=0}^{i-2} \frac{w - \mathcal{L} \underline{\delta}}{1 + K} \underline{\delta}^{p+1} \left(1 - \frac{w - \mathcal{L}}{1 + K} \right)^p \right] V^*(x) \\
&= \underline{\sigma}_i V^*(x). \quad (75)
\end{aligned}$$

Due to $w < 1 + K$, we have

$$0 < \lim_{i \rightarrow \infty} \underline{\sigma}_i = \underline{\delta} \frac{w(1 - \underline{\delta}) + \underline{\delta}(w - \mathcal{L})}{(1 + K)(1 - \underline{\delta}) + \underline{\delta}(w - \mathcal{L})} < 1. \quad (76)$$

Following the similar idea of the proof related to the lower bound of (70), it can be derived that if $\bar{\delta} < (1 + K)/(1 + K - w + \mathcal{L})$, then

$$\lim_{i \rightarrow \infty} \bar{\sigma}_i = \bar{\delta} \left(1 + \frac{(\bar{\delta} - 1)(1 + K - w)}{(1 + K) - \bar{\delta}(1 + K - w + \mathcal{L})} \right) > 1. \quad (77)$$

The proof is completed.

5. Simulation

Three examples are presented to verify the effectiveness of the developed method. In Example 1, a linear system is employed to show the capabilities of the algorithm, which is accelerating the convergence process and the impact of approximation error on convergence. In Example 2, a missile autopilot with nonlinear dynamics is applied to validate the availability of the established scheme for systems in the real world. Then, a nonlinear system with the uncertainty is given to further show the robustness of the obtained controller by using the constructed method.

5.1. Example 1

Consider a linear system written as

$$\dot{x} = \begin{bmatrix} 0.5 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} -2 \\ 1 \end{bmatrix} u, \quad (78)$$

where $x = [x_1, x_2]^T \in \mathbb{R}^2$ denotes the system state and $u(x) \in \mathbb{R}$ denotes the control input. For the utility function, Q and R are set as $Q = I_2$ and $R = 1$. The optimal control law can be solved as $u^*(x) = 1.2997x_1 - 0.0907x_2$. To verify [Theorems 5](#) and [6](#), two cases are presented in the following description.

A. Case 1

Considering the approximation error, the discrepancy between the estimated and true value functions is set as 0.1 and 0.001 to denote the cases that the condition in [Theorems 5](#) and [6](#) is satisfied or not, respectively. Since there are two cases related to the value of the relaxation factor, the case of $0 < w < 1$ is first discussed. When the difference between the estimated value function and the actual one is set as 0.001, the condition in [Theorem 5](#) is met leading to the stable state and the

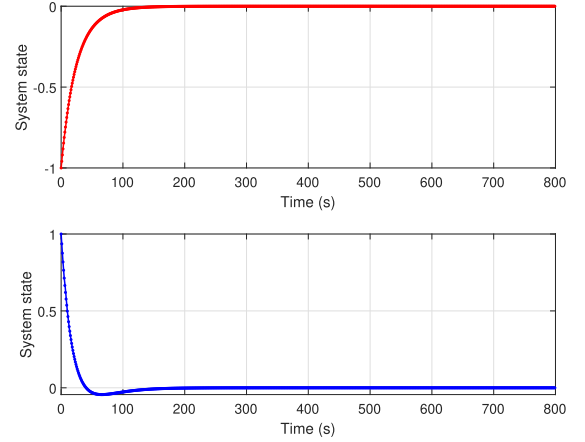


Fig. 1. State curves for the linear system with $0 < w < 1$.

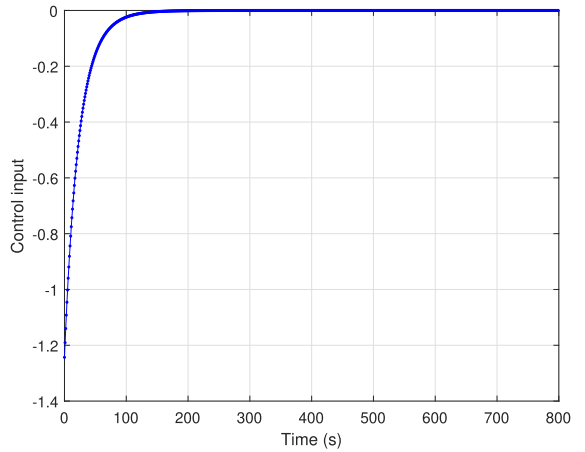


Fig. 2. Control curve for the linear system with $0 < w < 1$.

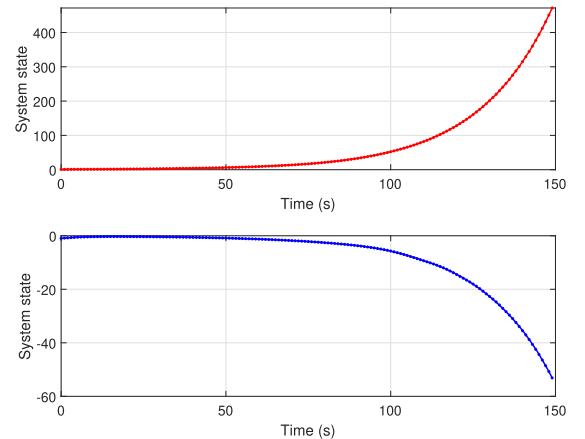
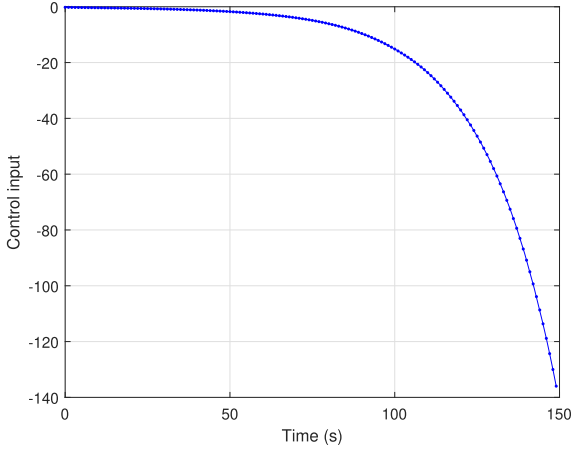
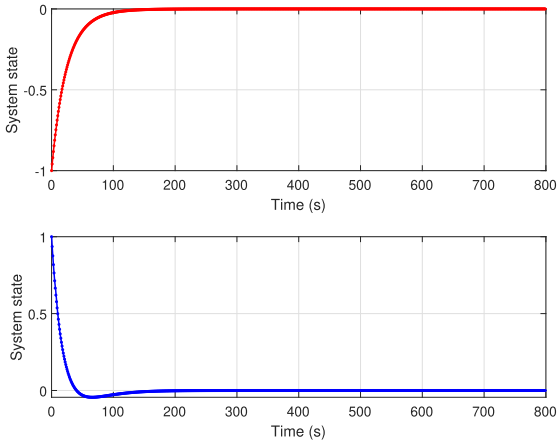


Fig. 3. State curves for the linear system with $0 < w < 1$.

control input converges to zero quickly, which can be seen in [Figs. 1](#) and [2](#). In contrast, as shown in [Figs. 3](#) and [4](#), the state of the system and the control input do not tend to zero when the difference is set to 0.1. Therefore, we can conclude that the analysis of the approximation error is reasonable and facilitate the implementation of the algorithm.

B. Case 2

Here, we analyze the case of $w > 1$. To satisfy the condition in [Theorem 6](#), the value of the difference is set as same as that in Case 1. According to [Figs. 5–8](#), the same results are presented to show that the

Fig. 4. Control curve for the linear system with $0 < w < 1$.Fig. 5. State curves for the linear system with $w > 1$.

system state and the control input converge to zero quickly when the condition is satisfied. However, without meeting the condition, the system remains unstable. To demonstrate the effect of speeding up the convergence process, three different values of w are selected to perform the developed algorithm, which are $w = 0.8$, $w = 1$, and $w = 1.3$. The norm of the weight vector $\|\hat{\omega}_c\|$ is selected as evaluation criteria. According to Fig. 9, the convergence process is presented with different relaxation factors. It can be seen that the convergence rate improves with a larger relaxation factor. From Fig. 9, the convergence results of NN weights are consistent. It implies that the learned near-optimal control policies are the same one corresponding to the consistent cost function.

5.2. Example 2

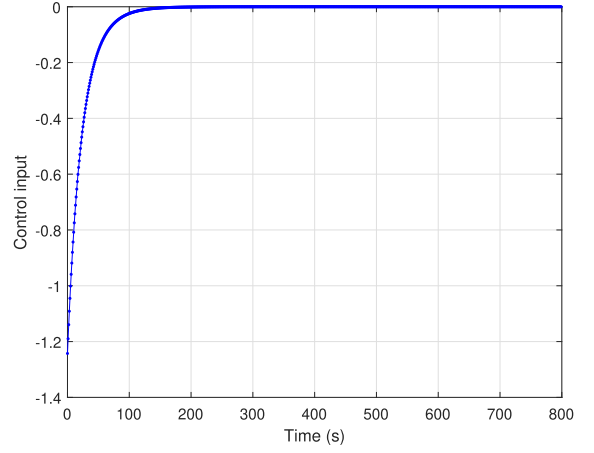
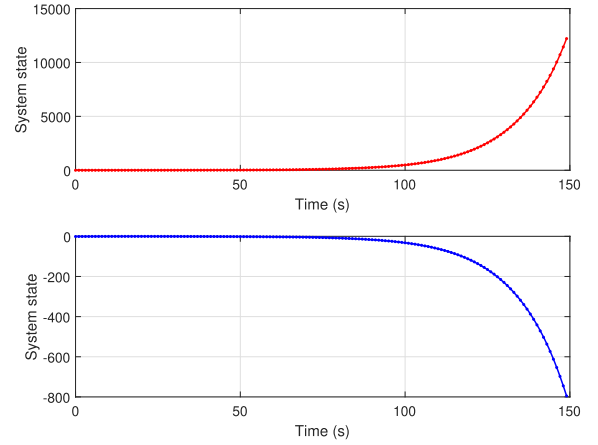
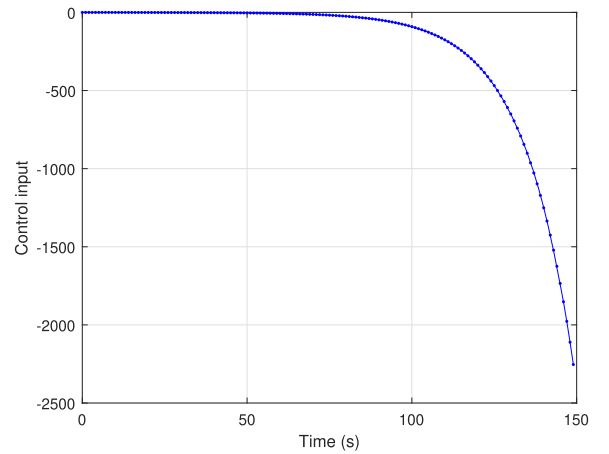
Consider the following missile autopilot with nonlinear dynamics:

$$\begin{cases} \dot{\alpha} = f_1(\alpha) + \beta + b_1(\alpha)\gamma \\ \dot{\beta} = f_2(\alpha) + b_2\gamma \\ \dot{\gamma} = \frac{-\gamma + u}{\mathcal{T}}, \end{cases} \quad (79)$$

where α , β , and γ denote the angle of attack, the pitch angular velocity, deflections of tail fins, respectively. $\mathcal{T}$ is a constant. The missile travels at Mach 3 at an altitude of 6095 m and the longitudinal dynamics is taken into consideration. The corresponding functions in (79) can be denoted as

$$f_1(\alpha) = \frac{18\mathcal{G}QS\alpha}{\pi Mv} (1.03 \times 10^{-3}\alpha^2 - 9.45 \times 10^{-2}|\alpha| - 1.7) \cos \frac{\pi\alpha}{180}, \quad (80)$$

$$f_2(\alpha) = \frac{180\mathcal{G}SD}{\pi M_p} (2.15 \times 10^{-4}\alpha^3 + 0.051\alpha - 1.95 \times 10^{-2}\alpha|\alpha|), \quad (81)$$

Fig. 6. Control curve for the linear system with $w > 1$.Fig. 7. State curves for the linear system with $w > 1$.Fig. 8. Control curve for the linear system with $w > 1$.

and

$$\begin{cases} b_1(\alpha) = -0.34 \frac{18\mathcal{G}QS}{\pi Mv} \cos \frac{\pi\alpha}{180} \\ b_2 = -2.06 \frac{180\mathcal{G}SD}{\pi M_p}, \end{cases} \quad (82)$$

where the corresponding parameters are shown in Table 1.

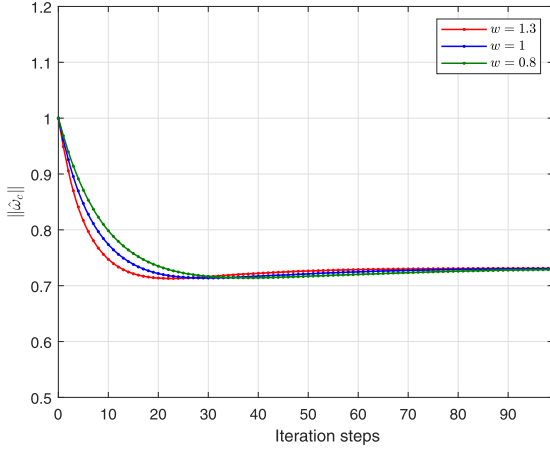


Fig. 9. The convergence process of the critic network weight norm for the non-linear system with different relaxation factors ($w = 0.8$, $w = 1$, and $w = 1.3$).

Table 1
Parameters of the missile autopilot.

Symbol	Meaning	Value
$\mathcal{M}/\text{kg}$	Mass of the missile autopilot	4410
$v/(\text{m} \cdot \text{s}^{-1})$	Velocity of the missile autopilot	947.6
$M_p/(\text{kg} \cdot \text{m}^2)$	The pitch moment	247
$q/(\text{N} \cdot \text{m}^{-2})$	Dynamic pressure	293
S/m^2	The reference area	0.04087
D/m	The reference diameter	0.229
$\mathcal{G}/(\text{m} \cdot \text{s}^{-2})$	Gravitational acceleration	9.8
$\mathcal{T}$	The time constant	0.1

Denote $x = [x_1, x_2, x_3]^T = [\alpha, \beta, \gamma]^T$. The system (79) can be represented as

$$\dot{x} = f(x) + g(x)u, \quad (83)$$

where

$$f(x) = \begin{pmatrix} f_1(x_1) + x_2 + b_1(x_1)x_3 \\ f_2(x_1) + b_2x_3 \\ -(1/\mathcal{T})x_3 \end{pmatrix}, g(x) = \begin{pmatrix} 0 \\ 0 \\ 1/\mathcal{T} \end{pmatrix}. \quad (84)$$

We set $Q = I_3$ and $R = 1$. The goal is to acquire the control input that can stabilize the system (79) and minimize the related cost function by using the developed algorithm. The activation function is set as $\sigma_c(x) = [x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2]^T$. To verify the effectiveness of the developed approach, we show the learning process of different values of w . As shown in Fig. 10, although the value of w is different, the final results all converge to the optimal weights related to the optimal value function. Besides, it can be found that the larger w , the faster the convergence speed. Using the obtained control law on the system (79), it can be seen that the system states reach the equilibrium point quickly according to Fig. 11. To further highlight the performance of accelerating the convergence process, computing time and iteration steps are presented in Table 2. It can be found that while the relaxation factor w primarily accelerates the offline learning phase without altering the equilibrium of the converged control policy, its practical value lies in the significant reduction of computational budget. Specifically, by employing a relaxation factor $w > 1$, the proposed algorithm achieves the convergence criterion within 40% fewer iterations compared to the standard VI ($w = 1$), leading to a substantial reduction in offline training time and computational resource allocation. Although the setting time under the near-optimal control law is similar, the computation burden is reduced by using the developed method.

Table 2
The comparison of performance indicators for Example 2.

Theorems	Computing time (s)	Iteration steps
Accelerated VI	153.8267	36
VI	255.3434	66

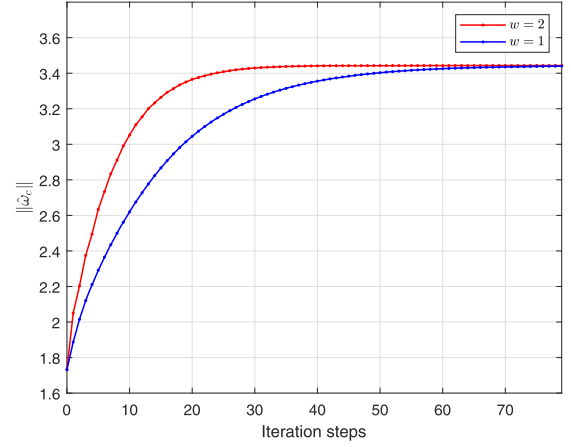


Fig. 10. The convergence process of the critic network weight norm for the missile autopilot system with different relaxation factors ($w = 1$ and $w = 2$).

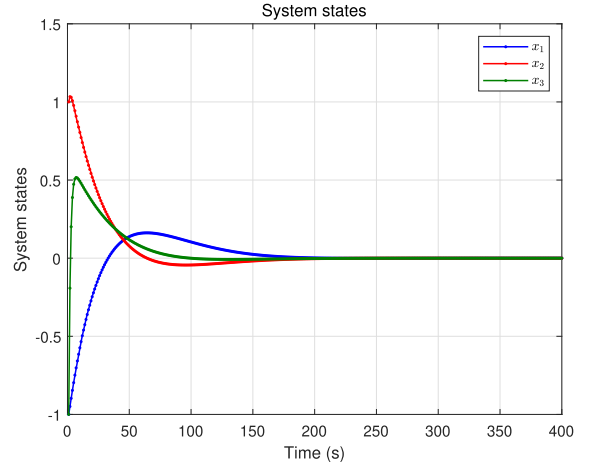


Fig. 11. State curves for the missile autopilot system.

5.3. Example 3

In Example 3, the robustness of the obtained control policy is demonstrated for the nonlinear system with the uncertainty.

Consider the following nonlinear system:

$$\dot{x} = \begin{bmatrix} -\sin(x_1) + x_2 \\ -2x_2 + 0.5x_2(\cos(2x_1) + 2)^2 \end{bmatrix} + \begin{bmatrix} x_2 \\ \cos(2x_1) + 2 \end{bmatrix} (u + 0.5\rho \sin x_2), \quad (85)$$

where $x = [x_1, x_2]^T \in \mathbb{R}^2$ denotes the system state, $u(x) \in \mathbb{R}$ denotes the control input, which is needed to be learned by using the constructed approach, and $d(x) = 0.5\rho \sin x_2$ is the matched uncertain term. Among that, ρ is the unknown parameter. Set $d_M(x) = \|x\|$ and $\rho = 1$. For the cost function, we choose $Q = I_2$ and $R = 1$. For the critic NN, the activation function is set as $\sigma_c(x) = [x_1^2, x_1x_2, x_2^2]^T$ and the weight vector is set as $\hat{w}_c = [\hat{w}_{c1}, \hat{w}_{c2}, \hat{w}_{c3}]^T$. The interval time is chosen as $T = 0.05\text{s}$. To verify the performance of the established scheme for nonlinear systems,

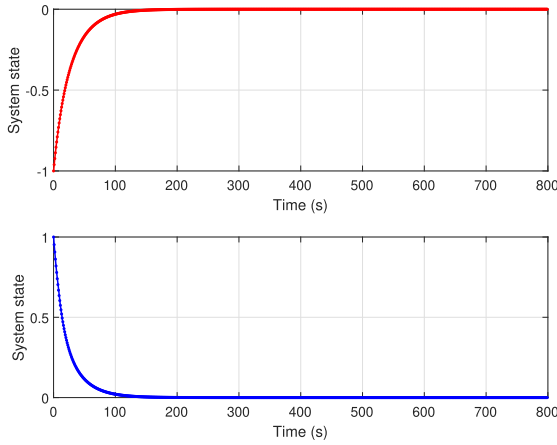


Fig. 12. State curves for the nonlinear system.

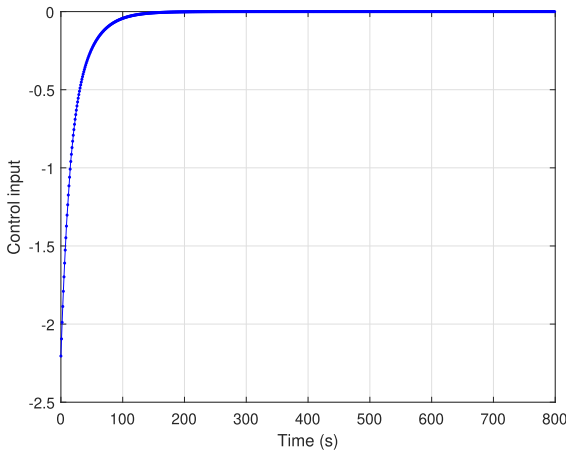


Fig. 13. Control curve for the nonlinear system.

Table 3

The comparison of performance indicators for Example 3.

Theorems	Computing time (s)	Iteration steps
Accelerated VI	20.8061	18
VI	25.6690	22

which is speeding the process of convergence, three different values are selected as $w = 0.75$, $w = 1$, and $w = 1.35$. According to Fig. 14, it can be found that the convergence rate increases with the value of w . It is proved that the developed approach is also valid for nonlinear systems. Then, the robustness of the established method is shown as Figs. 12 and 13. The obtained control law is utilized to control the system (85) after employing the advanced VI algorithm. Observing Figs. 12 and 13, it can be found that the system state and control input converge to zero under the learned control policy. It implies that the objective is accomplished, which is stabilizing the nonlinear system with the uncertainty. Similar to Example 2, computing time and iteration steps of accelerated VI ($w = 1.3$) and VI ($w = 1$) are presented in Table 3.

6. Conclusion

In conclusion, an advanced VI-based ADP mechanism is developed for CT systems with the uncertainty, focusing on its application and theoretical analysis. Significant advances in optimizing efficiency demonstrated by using the established mechanism. The analysis of the approximation error is presented to provide insights into the feasibility of the constructed mechanism that facilitates the practical application in en-

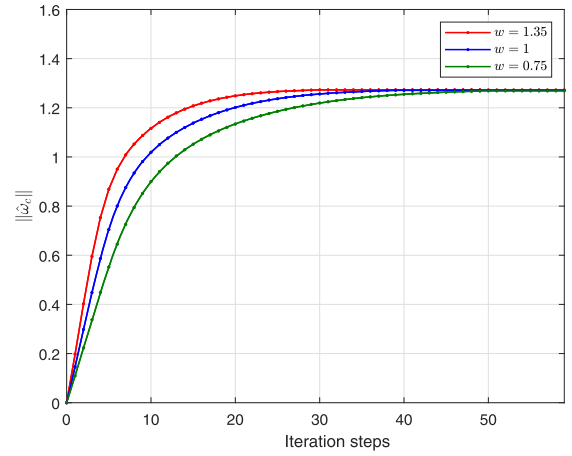


Fig. 14. The convergence process of the critic network weight norm for the nonlinear system with different relaxation factors ($w = 0.75$, $w = 1$, and $w = 1.35$).

gineering. Then, three examples are presented to show practical implementation is facilitated by analyzing the approximation error. In addition, effectiveness of the advanced VI-based ADP mechanism is demonstrated when handling the robust control problem of CT systems, particularly in improving convergence rates compared to traditional methods. However, the system model should be known, which leads to the limited application range of the algorithm. There are still numerous complex systems. Therefore, more learning mechanisms such as deep learning could be explored to expand the scalability of the developed algorithm. Furthermore, the optimal control law is learned offline by using the developed method, which leads to the same convergence results as traditional VI. To consider the performance of the actual controlled system, online learning should be investigated in the future.

Data availability

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

CRediT authorship contribution statement

Ao Liu: Writing – review & editing, Writing – original draft, Software, Methodology; **Ding Wang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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